

Hyper2 Test

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Hyper2 model for home advantage

Draws are excluded, so the model only compares matches with a winning team.

```
library(readxl)
library(dplyr)
library(hyper2)
library(tidyr)
library(ggplot2)
```

```
options(scipen = 999)
```

Load and combine SPFL datasets

```
# Function to load one SPFL season.
# file = Excel file name.
# season = season label to add to the dataset.
# sheet = worksheet name. If left blank, R reads the first sheet.

load_spfl <- function(file, season, sheet = NULL) {

  # Read the Excel file.
  # If no sheet is given, read the first sheet by default.
  if (is.null(sheet)) {
    data <- read_excel(file)
  } else {
    data <- read_excel(file, sheet = sheet)
  }

  # The pipe operator %>% takes the dataset on the left
# and passes it into the next function.
# This keeps only the columns needed for the model
# and then adds a Season column.
  data %>%
    select(`Home Team`, `Away Team`, Result) %>%
    mutate(Season = season)
}

# Load each season using the function above.
# Each dataset is reduced to Home Team, Away Team, Result, and Season.

spfl_2016_2017 <- load_spfl("SPFL 2016-2017.xlsx", "2016/17", "SPFL 2016-2017")
spfl_2017_2018 <- load_spfl("SPFL 2017-2018.xlsx", "2017/18", "SPFL 2017-2018")
```

```

spfl_2018_2019 <- load_spfl("SPFL 2018-2019.xlsx", "2018/19", "SPFL 2018-2019")
spfl_2019_2020 <- load_spfl("SPFL 2019-2020.xlsx", "2019/20", "SPFL 2019-2020")
spfl_2020_2021 <- load_spfl("SPFL 2020-2021.xlsx", "2020/21", "SPFL 2020-2021")
spfl_2021_2022 <- load_spfl("SPFL 2021-2022.xlsx", "2021/22")
spfl_2022_2023 <- load_spfl("SPFL 2022-2023.xlsx", "2022/23")
spfl_2023_2024 <- load_spfl("SPFL 2023-2024.xlsx", "2023/24")
spfl_2024_2025 <- load_spfl("SPFL 2024-2025.xlsx", "2024/25")
spfl_2025_2026 <- load_spfl("SPFL 2025-2026.xlsx", "2025/26", "SPFL Dataset")

# Combine each dataset into one larger dataset.

all_matches <- bind_rows(
  spfl_2016_2017, spfl_2017_2018, spfl_2018_2019, spfl_2019_2020, spfl_2020_2021,
  spfl_2021_2022, spfl_2022_2023, spfl_2023_2024, spfl_2024_2025, spfl_2025_2026
)

```

Choose dataset for the hyper2 model

The object `spfl_decided` controls which season is analysed. To analyse a different season, I change `spfl_2024_2025` to another season, such as `spfl_2020_2021`, or `all_matches`.

```

# Keep only matches with a winner.
# Draws are excluded because this Bradley-Terry style model compares winners and losers.

spfl_decided <- all_matches %>%
  filter(Result %in% c("Home", "Away"))

# Get all team names in the selected dataset.
# This accounts for different teams appearing in different seasons.

teams <- sort(unique(c(spfl_decided$`Home Team`, spfl_decided$`Away Team`)))

```

Build the hyper2 likelihood models

Two models are fitted. The first model estimates team strength only. The second model estimates team strength plus an additional `HomeAdvantage` parameter.

```

# Model 1: team strength only, with no home advantage parameter.
# Code was inspired from https://journal.r-project.org/articles/RJ-2017-061/

H_no_home <- hyper2(pnames = teams)

# Model 2: team strength plus a home advantage parameter.
# The HomeAdvantage parameter is treated as an additional strength component
# added to the home team in the Bradley-Terry likelihood.

H_home <- hyper2(pnames = c(teams, "HomeAdvantage"))

# Build both likelihoods match by match.
# For a match with a winner, the Bradley-Terry style likelihood is:
#
# winner strength / total strength of competitors
#
# For example, without home advantage:
# Home win = home / (home + away)

```

```

#
# With home advantage:
# Home win = (home + HomeAdvantage) / (home + HomeAdvantage + away)
# This part was aided by Chatgpt
# prompt was; "Use this code: "copy pasted all previous code", to help build 2 models
# 1 with and 1 without a home advantage component
# using the formulae provided by hyper2 (so in this case, home/home+away)
# and home + homeadv/ home+homeadv+away)"
# Using addition variant and not multiplication for simplicity - confirmed from supervisor
for (i in 1:nrow(spfl_decided)) {

  home <- spfl_decided$`Home Team`[i]
  away <- spfl_decided$`Away Team`[i]
  result <- spfl_decided$Result[i]

  if (result == "Home") {

    # No home advantage model:
    # Probability = home / (home + away)
    # Adding +1 puts the winner in the numerator.
    # Adding -1 to the pair puts both competitors in the denominator.

    H_no_home[home] <- H_no_home[home] + 1
    H_no_home[c(home, away)] <- H_no_home[c(home, away)] - 1

    # Home advantage model:
    # Probability = (home + HomeAdvantage) /
    #               (home + HomeAdvantage + away)

    H_home[c(home, "HomeAdvantage")] <- H_home[c(home, "HomeAdvantage")] + 1
    H_home[c(home, "HomeAdvantage", away)] <- H_home[c(home, "HomeAdvantage", away)] - 1

  } else if (result == "Away") {

    # No home advantage model:
    # Probability = away / (home + away)

    H_no_home[away] <- H_no_home[away] + 1
    H_no_home[c(home, away)] <- H_no_home[c(home, away)] - 1

    # Home advantage model:
    # Probability = away / (home + HomeAdvantage + away)

    H_home[away] <- H_home[away] + 1
    H_home[c(home, "HomeAdvantage", away)] <- H_home[c(home, "HomeAdvantage", away)] - 1

  }
}

```

Estimate strengths and compare models

```

# Estimate team strengths.
# mapp() is the hyper2 function which estimates the parameters by maximising the likelihood.
# Code inspired from https://journal.r-project.org/articles/RJ-2017-061/

```

```

mle_no_home <- maxp(H_no_home)
mle_home <- maxp(H_home)

# Show the estimates from each model.
mle_no_home

##      Aberdeen      Celtic      Dundee      Dundee United      Falkirk
##      0.05283484      0.33694101      0.01783051      0.02991613      0.03951019
##      Hamilton      Hearts      Hibernian      Inverness CT      Kilmarnock
##      0.01487372      0.05588662      0.05330870      0.01295772      0.03511813
##      Livingston      Motherwell      Partick Thistle      Rangers      Ross County
##      0.02263584      0.03349499      0.01871131      0.20243205      0.01953319
##      St Johnstone      St Mirren
##      0.02676587      0.02724918

mle_home

##      Aberdeen      Celtic      Dundee      Dundee United      Falkirk
##      0.051233840      0.357161528      0.015322757      0.026132856      0.038711523
##      Hamilton      Hearts      Hibernian      Inverness CT      Kilmarnock
##      0.012186687      0.053403906      0.052263096      0.008666659      0.031161843
##      Livingston      Motherwell      Partick Thistle      Rangers      Ross County
##      0.018058008      0.031383051      0.014787668      0.211554150      0.016760706
##      St Johnstone      St Mirren      HomeAdvantage
##      0.025403883      0.024554829      0.011253009

# Calculate log-likelihoods for both models.

ll_no_home <- loglik(mle_no_home, H_no_home)
ll_home <- loglik(mle_home, H_home)

ll_no_home

## [1] -935.3914

ll_home

## [1] -917.0083

# Likelihood ratio test.
# This compares the simpler no-home-advantage model
# against the model containing the HomeAdvantage parameter.

LRT <- 2 * (ll_home - ll_no_home)

# The LRT should not be negative in theory.
# This line avoids tiny negative values caused by numerical optimisation.

LRT <- max(0, LRT)

# The home advantage model adds one extra parameter,
# so the test uses 1 degree of freedom.

p_value <- pchisq(LRT, df = 1, lower.tail = FALSE)

LRT

```

```
## [1] 36.76616
p_value

## [1] 0.00000000133182
# Home advantage estimate.

mle_home["HomeAdvantage"]

## HomeAdvantage
##      0.01125301
# Main model log-likelihoods.
# Both are put into data frames just for easier summarising

hyper2_results <- data.frame(
  Model = c("No home advantage", "With home advantage"),
  LogLikelihood = c(ll_no_home, ll_home)
)

hyper2_results

##           Model LogLikelihood
## 1 No home advantage      -935.3914
## 2 With home advantage      -917.0083
# Final likelihood ratio test result.

lrt_results <- data.frame(
  LRT = LRT,
  PValue = p_value,
  HomeAdvantageEstimate = as.numeric(mle_home["HomeAdvantage"])
)

lrt_results

##           LRT           PValue HomeAdvantageEstimate
## 1 36.76616 0.00000000133182           0.01125301
```

Summary of hyper2 results across seasons

The following table records the results obtained by applying the same hyper2 model to each season individually and to all seasons combined.

```
# Create a summary table containing the hyper2 results for each season.
# These values were obtained by running the same model separately
# for each individual season and once for all seasons combined.
hyper2_summary_table <- data.frame(
  Season = c(
    "2016/17", "2017/18", "2018/19", "2019/20", "2020/21",
    "2021/22", "2022/23", "2023/24", "2024/25", "2025/26",
    "All seasons combined"
  ),
  # Log-likelihood values for the model without a home-advantage parameter.
  LogLik_NoHomeAdvantage = c(
    -78.87413, -88.22198, -89.98702, -67.00648, -78.44786,
```

```

-80.44761, -95.79790, -83.10915, -101.94840, -83.09382,
-935.26660
),

# Log-likelihood values for the model containing the HomeAdvantage parameter.
LogLik_WithHomeAdvantage = c(
-77.13478, -87.47995, -89.64439, -64.30464, -78.47971,
-77.29277, -89.46608, -79.33752, -99.31951, -80.51791,
-916.91210
),

# Likelihood ratio test statistics comparing the two models.
LRT = c(
3.478705, 1.484053, 0.6852672, 5.40368, 0,
6.309675, 12.5874, 7.543259, 5.25786, 5.151834,
36.70909
),

# p-values from the likelihood ratio tests.
PValue = c(
0.06216335, 0.2231415, 0.4077789, 0.02009434, 1,
0.01200809, 0.0003883527, 0.0060235, 0.02184779, 0.02322183,
1.371382e-09
),

# Estimated value of the HomeAdvantage parameter from the model.
HomeAdvantageEstimate = c(
4.399408e-06, 0.00553088, 0.004038126, 0.01451093, 0.000001,
0.009031458, 0.01563822, 0.01358, 0.01600478, 0.01387932,
0.0111028
),

# Written interpretation of whether each season showed evidence
# of home advantage based on the p-value.
Conclusion = c(
"Borderline, not significant", "Not significant", "Not significant",
"Significant", "No evidence", "Significant", "Strongly significant",
"Significant", "Significant", "Significant", "Very strong evidence"
)
)

# Display the final hyper2 summary table.
hyper2_summary_table

```

##	Season	LogLik_NoHomeAdvantage	LogLik_WithHomeAdvantage
## 1	2016/17	-78.87413	-77.13478
## 2	2017/18	-88.22198	-87.47995
## 3	2018/19	-89.98702	-89.64439
## 4	2019/20	-67.00648	-64.30464
## 5	2020/21	-78.44786	-78.47971
## 6	2021/22	-80.44761	-77.29277
## 7	2022/23	-95.79790	-89.46608
## 8	2023/24	-83.10915	-79.33752
## 9	2024/25	-101.94840	-99.31951

## 10	2025/26	-83.09382	-80.51791
## 11	All seasons combined	-935.26660	-916.91210
##	LRT	PValue	HomeAdvantageEstimate
## 1	3.4787050	0.0621633500000000	0.000004399408
## 2	1.4840530	0.2231415000000000	0.005530880000
## 3	0.6852672	0.4077789000000000	0.004038126000
## 4	5.4036800	0.0200943400000000	0.014510930000
## 5	0.0000000	1.0000000000000000	0.000001000000
## 6	6.3096750	0.0120080900000000	0.009031458000
## 7	12.5874000	0.0003883527000000	0.015638220000
## 8	7.5432590	0.0060235000000000	0.013580000000
## 9	5.2578600	0.0218477900000000	0.016004780000
## 10	5.1518340	0.0232218300000000	0.013879320000
## 11	36.7090900	0.000000001371382	0.011102800000
##	Conclusion		
## 1	Borderline, not significant		
## 2	Not significant		
## 3	Not significant		
## 4	Significant		
## 5	No evidence		
## 6	Significant		
## 7	Strongly significant		
## 8	Significant		
## 9	Significant		
## 10	Significant		
## 11	Very strong evidence		

Interpretation

The hyper2 results show that the model including a **HomeAdvantage** parameter fitted significantly better than the no-home-advantage model for several seasons, including 2019/20, 2021/22, 2022/23, 2023/24, 2024/25 and 2025/26. The all-seasons combined model also provided very strong evidence of home advantage.

In contrast, the 2020/21 COVID-affected season showed no evidence of home advantage, with the estimated **HomeAdvantage** parameter being essentially zero. This supports the idea that home advantage was weaker or absent during the COVID-affected season, although the model excludes draws and should therefore be interpreted as an initial likelihood-based analysis rather than a complete model of all match outcomes.

Plot

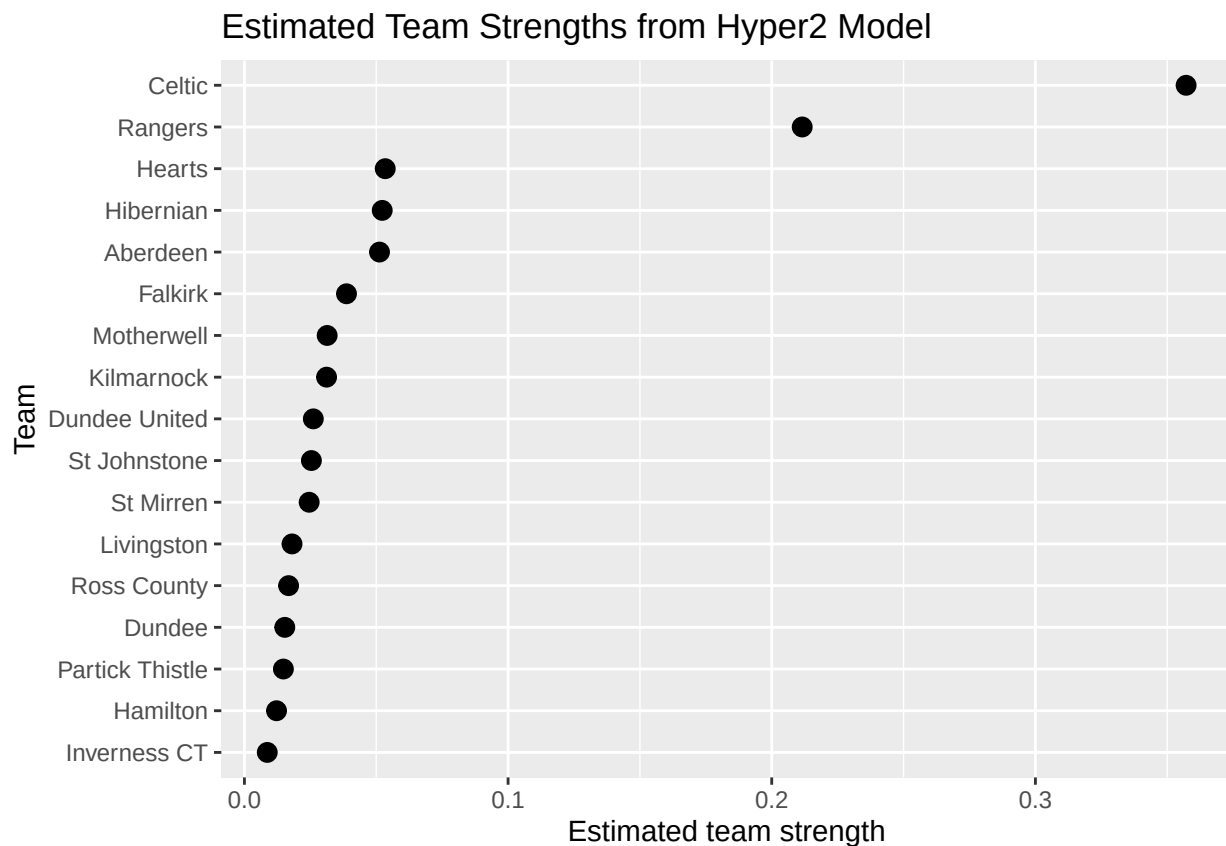
```
# Convert the MLE estimates from the home-advantage model into a data frame.
# This allows the estimated team strengths to be plotted using ggplot.
team_strength_estimates <- data.frame(
  Team = names(mle_home),
  StrengthEstimate = as.numeric(mle_home)
) %>%

# Remove the HomeAdvantage parameter because this plot only shows teams.
filter(Team != "HomeAdvantage") %>%

# Order teams by estimated strength.
arrange(StrengthEstimate)

# Produce a dot plot of estimated team strengths from the hyper2 model.
```

```
ggplot(team_strength_estimates,
      aes(x = StrengthEstimate,
          y = reorder(Team, StrengthEstimate))) +
geom_point(size = 3) +
labs(
  title = "Estimated Team Strengths from Hyper2 Model",
  x = "Estimated team strength",
  y = "Team"
)
```



```
# Remove the all-seasons combined row so that the plot only compares
# individual seasons.
hyper2_season_table <- hyper2_summary_table %>%
  filter(Season != "All seasons combined")

# Calculate the 5% critical value for a chi-square distribution
# with 1 degree of freedom.
critical_value <- qchisq(0.95, df = 1)

# Produce a bar chart of the likelihood ratio test statistic by season.
ggplot(hyper2_season_table,
      aes(x = Season, y = LRT)) +
geom_col() +

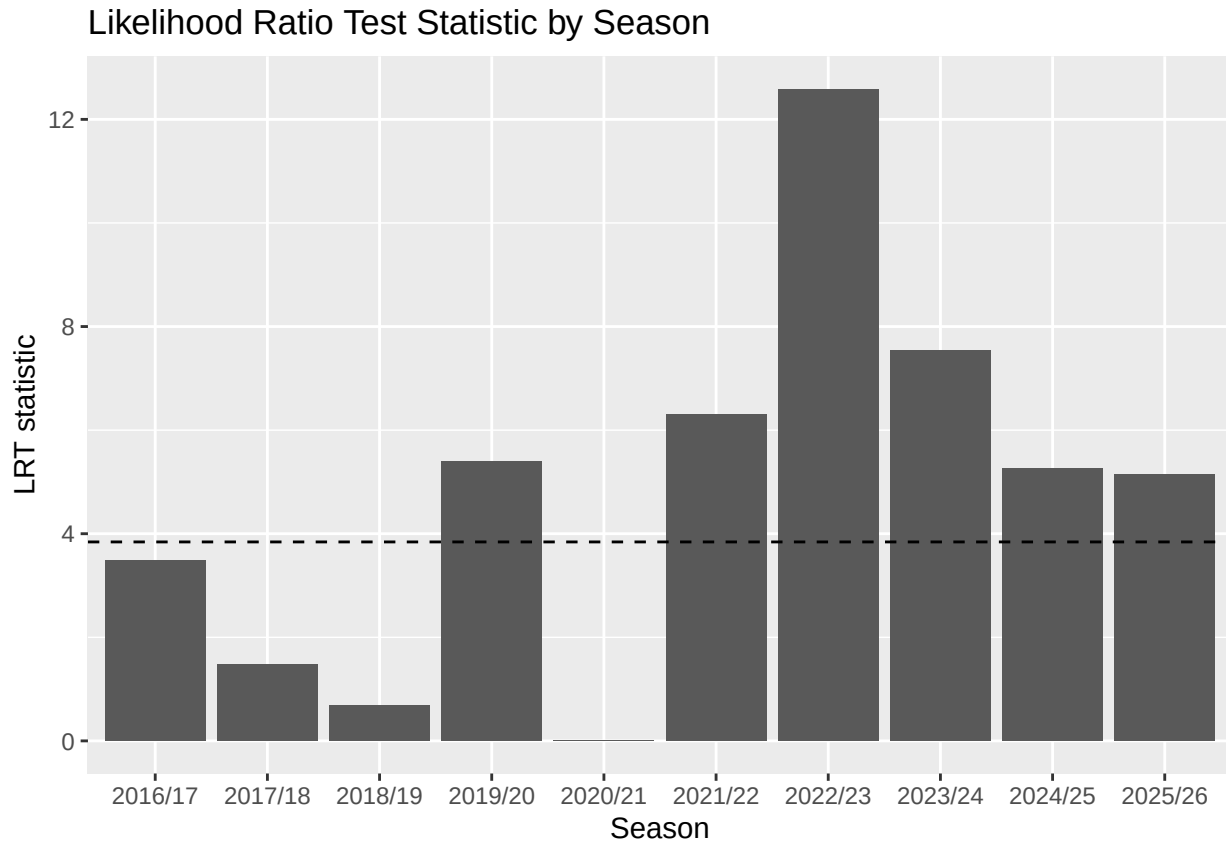
# Add a dashed reference line showing the 5% significance threshold.
geom_hline(yintercept = critical_value, linetype = "dashed") +
labs(
```



```

title = "Likelihood Ratio Test Statistic by Season",
x = "Season",
y = "LRT statistic"
)

```

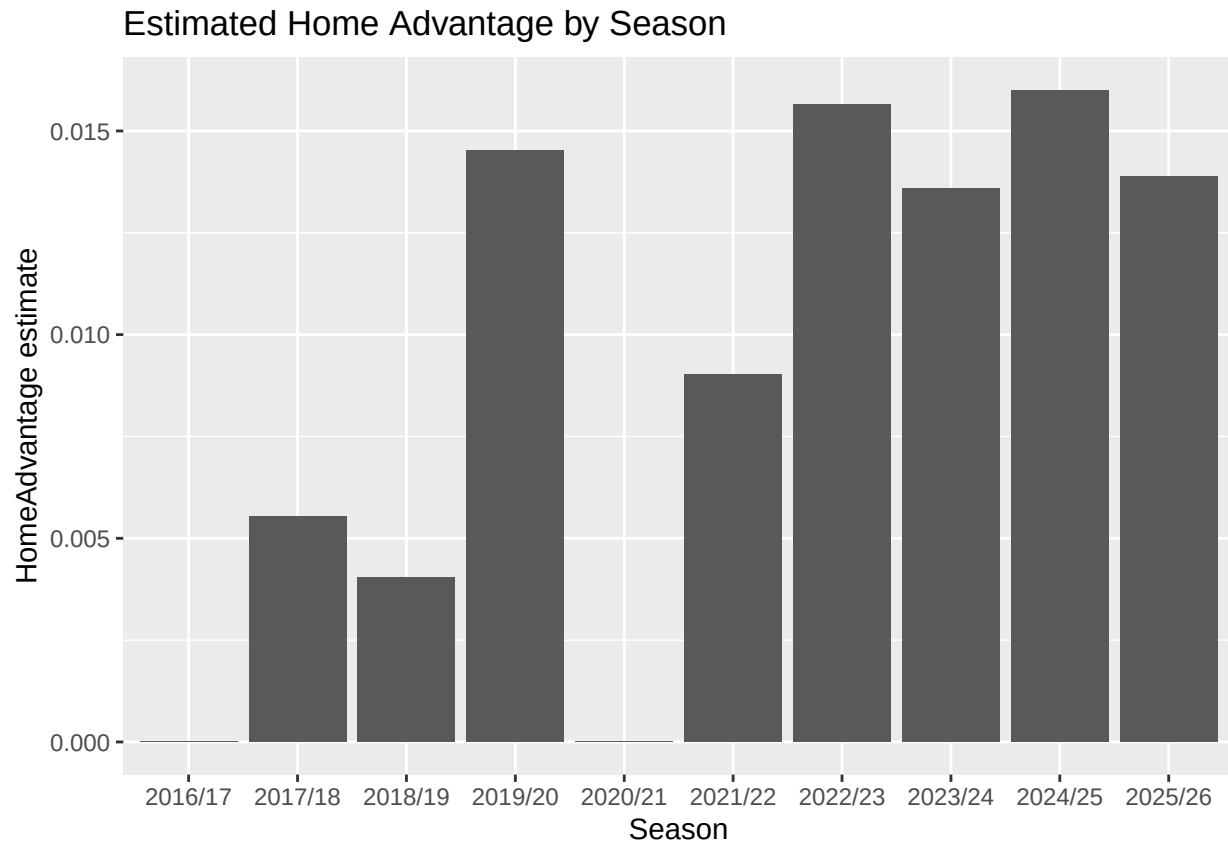


The dashed horizontal line represents the approximate 5% critical value for a chi-square distribution with one degree of freedom. Seasons above this line provide statistically significant evidence that the home-advantage model fitted better than the no-home-advantage model. The 2020/21 season was clearly below this threshold, while the seasons from 2021/22 to 2025/26 were above it.

```

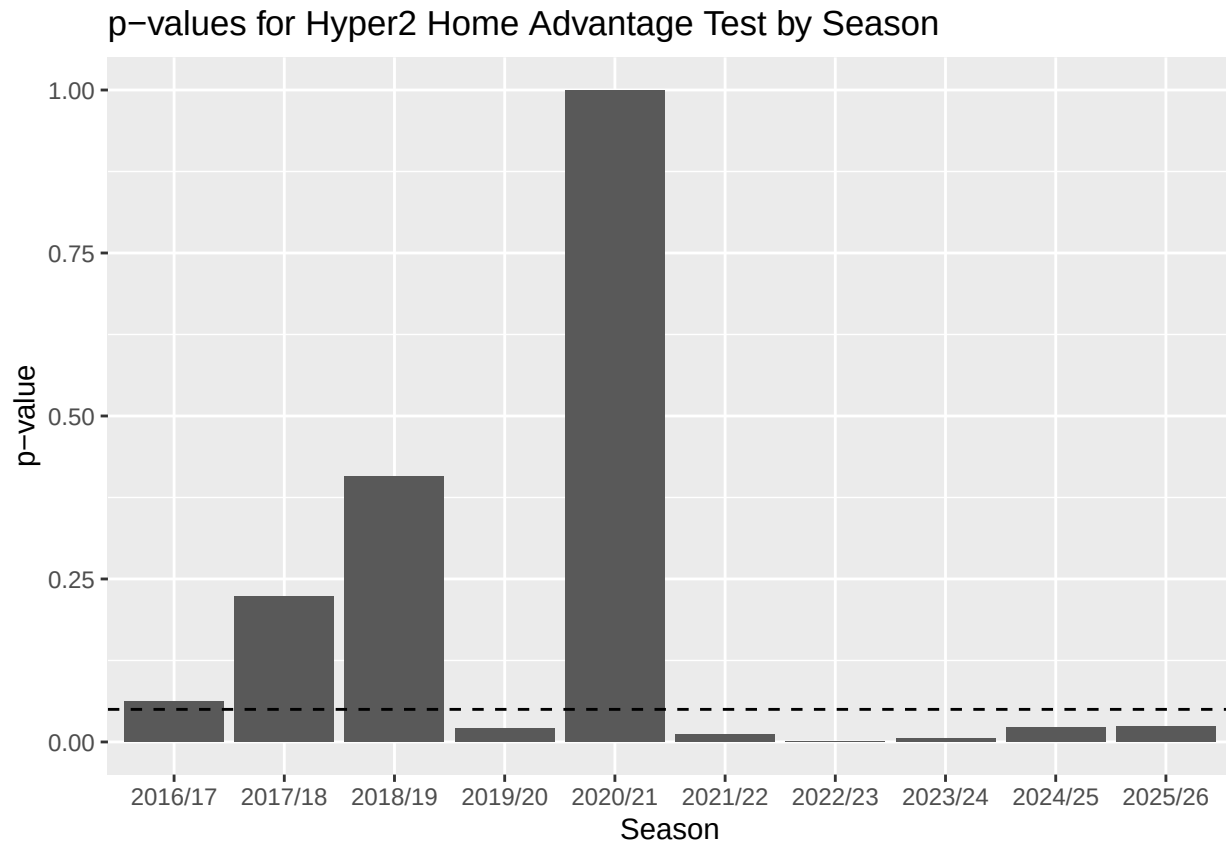
# Plot the estimated HomeAdvantage parameter for each individual season.
# Higher values indicate a stronger estimated home-advantage effect.
ggplot(hyper2_season_table,
  aes(x = Season, y = HomeAdvantageEstimate)) +
  geom_col() +
  labs(
    title = "Estimated Home Advantage by Season",
    x = "Season",
    y = "HomeAdvantage estimate"
  )
)

```



```
# Produce a bar chart showing the p-value from the likelihood ratio test
# for each season.
ggplot(hyper2_season_table,
       aes(x = Season, y = PValue)) +
  geom_col() +

  # Add a dashed line at p = 0.05 to show the usual significance level.
  geom_hline(yintercept = 0.05, linetype = "dashed") +
  labs(
    title = "p-values for Hyper2 Home Advantage Test by Season",
    x = "Season",
    y = "p-value"
  )
```

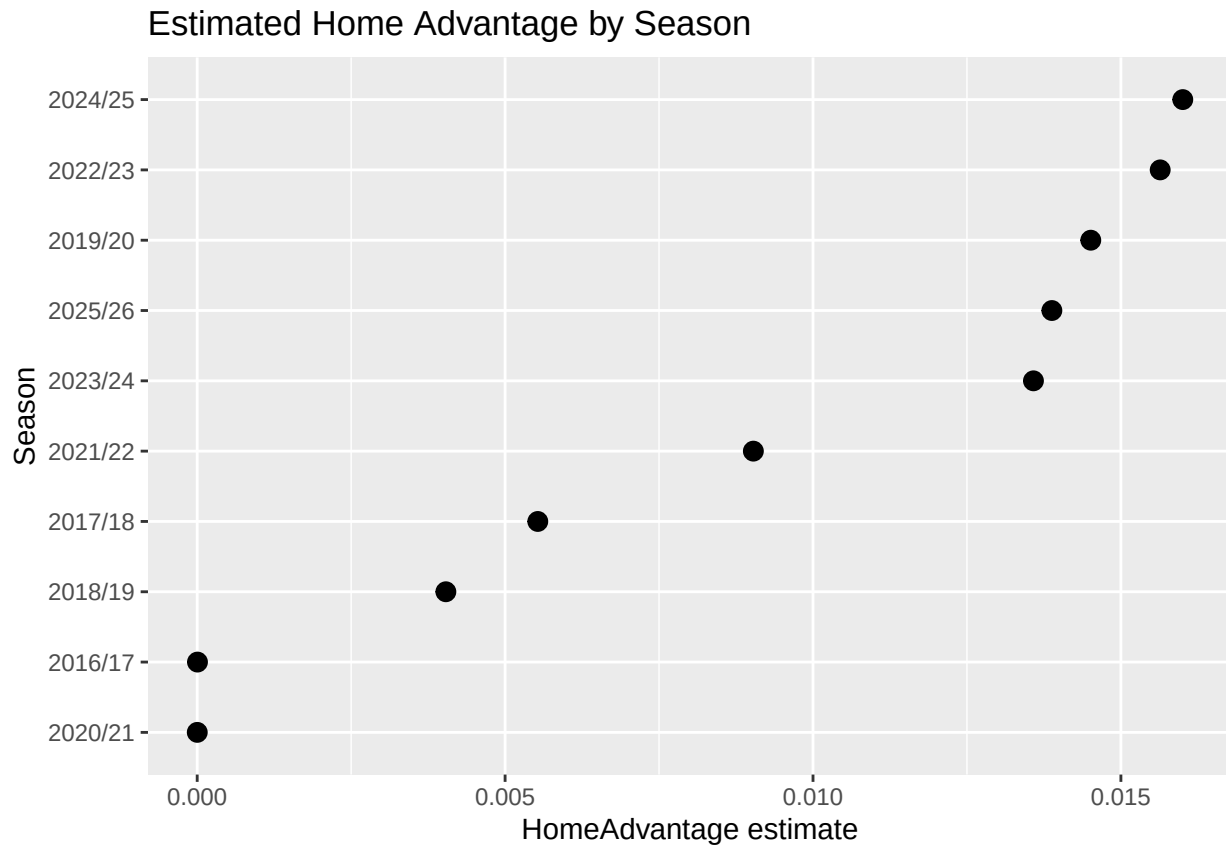


The p-value plot shows the p-values from the likelihood ratio tests comparing the no-home-advantage hyper2 model with the home-advantage model for each season. The dashed horizontal line represents the 5% significance level.

The results show that 2019/20 and every season from 2021/22 to 2025/26 had statistically significant evidence of home advantage. In contrast, the 2020/21 COVID-affected season had a p-value of 1, indicating no evidence that the home-advantage model improved fit. This supports the view that home advantage was weaker or absent during the COVID-affected season.

Does champion dominance affect home advantage parameter?

```
# Produce a dot chart of HomeAdvantage estimates by season.
# This provides another way to compare the estimated strength of home advantage
# across seasons.
ggplot(hyper2_season_table,
  aes(x = HomeAdvantageEstimate,
    y = reorder(Season, HomeAdvantageEstimate))) +
geom_point(size = 3) +
labs(
  title = "Estimated Home Advantage by Season",
  x = "HomeAdvantage estimate",
  y = "Season"
)
```



Display the summary table again so that it can be viewed alongside the plot.
hyper2_summary_table

##	Season	LogLik_NoHomeAdvantage	LogLik_WithHomeAdvantage
## 1	2016/17	-78.87413	-77.13478
## 2	2017/18	-88.22198	-87.47995
## 3	2018/19	-89.98702	-89.64439
## 4	2019/20	-67.00648	-64.30464
## 5	2020/21	-78.44786	-78.47971
## 6	2021/22	-80.44761	-77.29277
## 7	2022/23	-95.79790	-89.46608
## 8	2023/24	-83.10915	-79.33752
## 9	2024/25	-101.94840	-99.31951
## 10	2025/26	-83.09382	-80.51791
## 11	All seasons combined	-935.26660	-916.91210
##	LRT	PValue	HomeAdvantageEstimate
## 1	3.4787050	0.0621633500000000	0.000004399408
## 2	1.4840530	0.2231415000000000	0.005530880000
## 3	0.6852672	0.4077789000000000	0.004038126000
## 4	5.4036800	0.0200943400000000	0.014510930000
## 5	0.0000000	1.0000000000000000	0.000001000000
## 6	6.3096750	0.0120080900000000	0.009031458000
## 7	12.5874000	0.0003883527000000	0.015638220000
## 8	7.5432590	0.0060235000000000	0.013580000000
## 9	5.2578600	0.0218477900000000	0.016004780000
## 10	5.1518340	0.0232218300000000	0.013879320000
## 11	36.7090900	0.000000001371382	0.011102800000

```
##                Conclusion
## 1  Borderline, not significant
## 2                Not significant
## 3                Not significant
## 4                Significant
## 5                No evidence
## 6                Significant
## 7      Strongly significant
## 8                Significant
## 9                Significant
## 10               Significant
## 11      Very strong evidence

# Create a table showing the points gap between the league champion
# and the second-placed team in each season.
champion_margin_table <- data.frame(
  Season = c(
    "2016/17", "2017/18", "2018/19", "2019/20", "2020/21",
    "2021/22", "2022/23", "2023/24", "2024/25", "2025/26"
  ),
  ChampionMargin = c(
    30, 9, 9, 13, 25, 4, 7, 8, 17, 2
  )
)

# Join the champion-margin data onto the hyper2 summary table.
# This allows the home-advantage estimate to be compared with how dominant
# the league champion was in each season.
home_advantage_vs_margin <- hyper2_summary_table %>%
  filter(Season != "All seasons combined") %>%
  left_join(champion_margin_table, by = "Season")

# Display the combined table.
home_advantage_vs_margin
```

##	Season	LogLik_NoHomeAdvantage	LogLik_WithHomeAdvantage	LRT
## 1	2016/17	-78.87413	-77.13478	3.4787050
## 2	2017/18	-88.22198	-87.47995	1.4840530
## 3	2018/19	-89.98702	-89.64439	0.6852672
## 4	2019/20	-67.00648	-64.30464	5.4036800
## 5	2020/21	-78.44786	-78.47971	0.0000000
## 6	2021/22	-80.44761	-77.29277	6.3096750
## 7	2022/23	-95.79790	-89.46608	12.5874000
## 8	2023/24	-83.10915	-79.33752	7.5432590
## 9	2024/25	-101.94840	-99.31951	5.2578600
## 10	2025/26	-83.09382	-80.51791	5.1518340

##	PValue	HomeAdvantageEstimate	Conclusion
## 1	0.0621633500	0.000004399408	Borderline, not significant
## 2	0.2231415000	0.005530880000	Not significant
## 3	0.4077789000	0.004038126000	Not significant
## 4	0.0200943400	0.014510930000	Significant
## 5	1.0000000000	0.000001000000	No evidence
## 6	0.0120080900	0.009031458000	Significant
## 7	0.0003883527	0.015638220000	Strongly significant
## 8	0.0060235000	0.013580000000	Significant

## 9	0.0218477900	0.016004780000	Significant
## 10	0.0232218300	0.013879320000	Significant
##	ChampionMargin		
## 1	30		
## 2	9		
## 3	9		
## 4	13		
## 5	25		
## 6	4		
## 7	7		
## 8	8		
## 9	17		
## 10	2		

```

# Produce a scatter plot comparing champion dominance with the estimated
# home-advantage parameter.
ggplot(home_advantage_vs_margin,
       aes(x = ChampionMargin,
           y = HomeAdvantageEstimate,
           label = Season)) +
  geom_point(size = 3) +

  # Add season labels next to each point.
  geom_text(vjust = -0.5, size = 3) +
  labs(
    title = "Champion Dominance Compared with Hyper2 Home Advantage",
    x = "Champion points margin",
    y = "Home advantage estimate"
  )

```

